

Seongwoo Choi

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EDUCATION

- University of Pennsylvania** Philadelphia, PA
• *Master of Science in Engineering in Artificial Intelligence; GPA: 4.0/4.0* August 2025 - August 2027 (expected)
Courses: Artificial Intelligence, Natural Language Processing, Machine Learning for Data Science, Statistics for Data Science, Principles of Deep Learning, Technology Ethics and the Legal Landscape, GPU Computing for Machine Learning Systems, Computer Vision & Computational Photography
- University of California, Davis** Davis, CA
• *Master of Science in Computer Science; GPA: 3.71/4.0* September 2020 - December 2022
Courses: Deep Learning, Advanced Computer Architectures, Design and Analysis Of Algorithms, Data Analysis in R and Web Technologies, Machine Learning and Discovery, Distributed Databases, Software Engineering, Advanced Deep Learning
- University of California, Santa Cruz** Santa Cruz, CA
• *Bachelor of Science in Computer Science; GPA: 3.25/4.0* September 2013 - June 2017
Courses: Operating Systems, Data Structures and Algorithms, Abstract Data Types, Intermediate Programming, Advanced Programming in C++, Analysis Of Algorithms, Web Programming, Programming Languages
- Stanford Graduate School of Business** Stanford, CA
• *Stanford LEAD Certificate Program (Executive Education) – Admitted* March 2027 - March 2029

SKILLS SUMMARY

- **Languages:** Python, SQL, Bash, TypeScript
- **ML/AI:** PyTorch, TensorFlow, VLM / OCR pipelines, evaluation
- **Frameworks:** NextJS, LangChain/LangGraph, FastAPI
- **Technologies:** Docker, Git, CI/CD, AWS EC2, Azure OpenAI Services, Microsoft Copilot Studio
- **Platforms:** macOS, Ubuntu Linux, Windows
- **Soft Skills:** Leadership, Writing, Public Speaking
- **Spoken Languages:** Korean(Native), English (Fluent), Mandarin (Advanced), Spanish (Elementary)

PROFESSIONAL EXPERIENCE

- Aewol Labs, Inc. (Delaware C-Corp)** Abu Dhabi, UAE / Remote
• *Founder* September 2026 – Present
 - Building AewolFlow, a source-grounded research workspace: turns a scientific paper into a short note in which every claim links to a highlighted passage in the original PDF, shareable by URL without an account.
 - Accepted into Anthropic's Claude for Startups program (Sep 2026).
- Astra Tech (a G42 company)** Abu Dhabi, UAE
• *Senior AI Engineer – Artificial Intelligence Business Unit* March 2026 – Present
 - Technical lead within a 35-person cross-functional team. Led end-to-end delivery of the MVP of BOTIM's AI customer-support chatbot (Mira) on a messaging and fintech platform serving approximately 150 million users.
 - Architected extensible micro-service and mini-app integration layers enabling low-latency model inference across multi-division enterprise operations.
 - Since July 2026, lead an internal agentic-AI workstream automating HR and business-operations workflows.
- LG CNS** Seoul, Republic of Korea
• *AI Research Engineer – Multi-Modal Intelligence Lab* September 2023 – March 2026
 - Engineered enterprise-grade generative AI and multimodal document intelligence solutions incorporating LLM, RAG, and VLM architectures.
 - Managed and deployed centralized generative AI software pipelines across 10+ operational enterprise teams.
 - Architected a high-throughput RAG pipeline leveraging Python, LangChain, and Azure OpenAI PTU for enterprise PDF intelligence extraction.
 - Designed a VLM parsing pipeline integrating DeepSeek OCR, Qwen-VL, and PaddleOCR-VL; engineered synthetic data generation pipelines to fine-tune production models.
 - Led an Azure OpenAI architecture demonstration to executive leadership from LG CNS and Microsoft Asia.
- Intel Corporation** Davis, CA
• *AI Research Engineer & GPU Solutions Architect (Graduate Intern)* June 2022 – October 2022
 - Researched novel Graph Neural Network (GNN) architectures on the AVA Dataset utilizing PyTorch Geometric.

- Evaluated cloud gaming GPU workloads using system performance profiling tools (Intel vTune Profiler, Intel SoC Watch) to diagnose hardware bottlenecks.

- **UC Davis Graduate Entrepreneurship Club** Davis, CA
Co-Founder & Machine Learning Engineer – vAnalytix *December 2021 – June 2022*
 - Developed cross-platform vehicle diagnostics applications utilizing TensorFlow to build predictive vehicle health models from OBD-II telemetry.
- **United States Army Garrison Korea** Pyeongtaek, South Korea
Korean Augmentation to the United States Army *October 2017 – June 2019*
 - Served twenty months of mandatory service.
 - Built data-analysis software used by the unit’s commander and staff; recognized with a Certificate of Appreciation from the garrison command (January 2019).

RESEARCH & OPEN-SOURCE LEADERSHIP

- **Apache ResilientDB (Incubating)** Davis, CA / Remote
Initial Committer & Podling PMC Member (since incubation, Oct 2023) *February 2022 – Present*
 - Named as an initial committer in the project’s Apache Incubator proposal; ResilientDB, a high-throughput, fault-tolerant distributed database, entered the Apache Incubator in October 2023.
 - As a graduate student researcher, recompiled the codebase for Linux and cloud environments and authored the MkDocs architecture documentation for NexRes (second-generation ResilientDB).
- **Davis Computer Architecture Research Team** Davis, CA
Graduate Research Assistant *March 2022 – June 2022*
 - Debugged and ported performance benchmark applications across ARM, RISC-V, and x86 hardware architectures.
- **Plug-In Hybrid Electric Vehicles Research Center** Davis, CA
Graduate Student Researcher *February 2021 – June 2022*
 - Conducted predictive modeling research on EV battery state-of-charge patterns, co-authoring academic manuscripts and geospatial pipelines.

PROFESSIONAL SERVICE & MEMBERSHIPS

- **Conference on Neural Information Processing Systems (NeurIPS)** Remote
Workshop Reviewer *2025, 2026*
 - Workshop reviewer, MTI-LLM @ NeurIPS 2025; Meta-Agents @ NeurIPS 2026
 - Invited to judge and review peer-submitted research papers for premier machine learning workshops.
- **Sigma Xi, The Scientific Research Honor Society** Global
Full Member *Elected July 2026*
 - Nominated and inducted as a Full Member in recognition of noteworthy original research contributions in distributed systems and applied machine learning.

SELECTED PUBLICATIONS & MANUSCRIPTS

- **Preprint/Under Review:** “Do KV Cache Compression Strategies Compose? An Empirical Study of Quantization and Token Eviction on Edge Hardware,” *Manuscript, 2026; to be submitted to the Tenth Annual Conference on Machine Learning and Systems (MLSys 2027)*, 2027.
- **Conference/ArXiv:** “Predictive Modeling of Charge Levels for Battery Electric Vehicles using CNN EfficientNet and IGTD Algorithm,” *arXiv:2206.03612*, 2022.
- **Research Manuscript:** “Influence of Communication Among Shared Developers on the Productivity of Open Source Software Projects,” *arXiv:2203.15459*, 2022.
- **Research Manuscript:** “A Deep Learning Technique using a Sequence of Follow Up X-Rays for Disease Classification,” *arXiv:2203.15060*, 2022.

HONORS, AWARDS & CERTIFICATIONS

- **Honors & Awards:** Best Contributor (UC Davis CS Entrepreneurship Club, 2022); Certificate of Appreciation (USAG-Pyeongtaek, 2019); Dean’s Honor Award (UC Santa Cruz, 2013–2017).
- **Industry Certifications:** NVIDIA Certified Associate (GenAI LLMs); Google TensorFlow Developer Certificate; Deep Learning Specialization (DeepLearning.AI); Machine Learning Specialization (Stanford Online).